

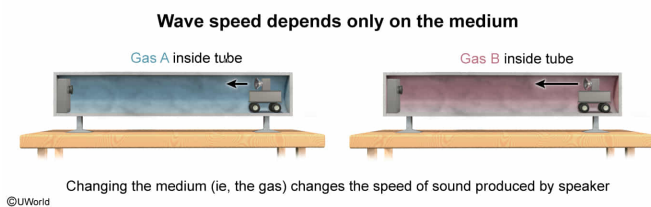
An apparatus consists of a cart that can move along a horizontal track inside a sealed tube that can be filled with various gases. Researchers attached a speaker to the cart, and a sensor that can measure sound is located at one end of the tube. Which of the following procedures will allow the researchers to change the speed of the sound created by the speaker?

- ☐ A. Set the cart in motion either toward or away from the sensor [11%]
- ☐ B. Change the frequency of the sound produced by the speaker [27%]
- ✓ ☒ C. Fill the tube with a different gas [47%]
- ☐ D. None of these actions will change the speed of the sound [13%]

Correct

47% answered correctly

Explanation:



A disturbance in a medium (eg, gas, water) creates a mechanical wave that propagates throughout the medium. When the properties of the medium are independent of **wavelength** λ , waves travel with a speed v that equals the product of λ and the **frequency** f of the wave:

$$v = \lambda f$$

However, the **wave speed** itself is determined only by the *physical properties* of the **medium**, such as temperature, elasticity, and density. Consequently, all waves in a medium travel at the same speed regardless of their frequency, wavelength, or period.

In this question, the researchers want to change the speed of the speaker sound. Because v depends only on the medium, they can only change v by changing the medium in which that sound propagates. Therefore, filling the tube with a different gas is the only procedure listed that would change the speed of sound v in the tube.

(Choice A) The **Doppler effect** implies that the motion of the cart changes

the frequency and wavelength of the sound received by the sensor.
However, wave speed is not affected by the motion of the source.

(Choice B) The speed of the sound is independent of its frequency.

(Choice D) The speed of a wave is not universal but rather depends on the medium of propagation.

Educational objective:

The speed of a wave depends only on the properties of the medium of propagation. Changing the medium (eg, changing the gas in a tube) changes the speed of sound in that medium.

Time spent:0 sec

Subject:

Physics

QID:403969

System:

Light and Sound